

WELL COMPLETION REPORT (SYNTHETIC)

OIL-KMC-006 - Kumchai-006

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000072
Field / District	Kumchai / Tinsukia
Block	OALP-IV-04
Basin	Assam-Arakan
Formation	Barail (BAR)
Spud / Completion	2024-05-30 / 2024-09-05
Measured / TVD	2583 m / 2549 m
Trajectory	Directional; inc 17.1 deg; az 178.4 deg
Formation Interval	1152-2122 m
Lithology	Interbedded sandstone and shale
Pore Pressure Reference	2602 psi
Fracture Gradient	14.92 ppg
Mud System	PHPA Water-Based Mud
NWIS Well Risk	Moderate (46.4/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	645	Surface
9 5/8 in casing	1498	Intermediate
7 in liner/casing	2273	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2024-06-10 430 m	Stuck Pipe	High	Rotary movement deteriorated progressively. Mitigation: Improve hole cleaning, work pipe and back-ream within operating envelope. Lesson: Avoid prolonged static periods in high-drag interval.
2024-06-14 541 m	Cementing Issue	Medium	Observed returns did not match calculated displacement. Mitigation: Review loss history before cementing equivalent interval. Lesson: Offset loss history should inform cementing hydraulics.
2024-06-18 649 m	Pressure Spike	Medium	Short-duration high-pressure excursion was recorded. Mitigation: Use pressure trend thresholds referenced to depth and flow rate. Lesson: Use depth-matched offset hydraulic baseline for early warning.
2024-06-29 913 m	Stuck Pipe	Medium	High drag was recorded while circulating remained possible. Mitigation: Condition mud and verify cuttings transport before connection. Lesson: Avoid prolonged static periods in high-drag interval.
2024-07-02 1004 m	Mud Loss	Medium	Partial returns observed at shale shaker. Mitigation: Spot graded LCM pill and circulate at reduced rate. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.
2024-07-03 1017 m	Mud Loss	Medium	Intermittent circulation losses developed while drilling ahead. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Track flow balance and pit volume trend before losses become severe.
2024-07-28 1644 m	Torque Spike	Medium	Torque rose without corresponding pressure anomaly. Mitigation: Monitor torque derivative rather than absolute torque alone. Lesson: Use offset torque trend to set adaptive operating envelope.

Completion Summary

The synthetic well reached 2583 m MD in the Barail interval. The NWIS knowledge record links 9 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.